

AnyCam 3.0.0-rc1.0 Audit Report

Release audit - 02 October 2026 - tests in the repository, module split stage 1, log redaction, dead endpoints - tag 3.0.0-rc1.0

3.0.0-rc1.0 is a code structure release: AnyCam's behaviour is meant to stay the same. Scope confirmed by CrystalHeeler on 2026-10-02: build plan E7, E6, E8, E4 and E1; E2, E3 and E5 dropped; version 3.0.0-rc1.0; stop and ask on problems. Built: E7, E6, E8, E4 and stage 1 of E1. Not built: E1 stages 2 and 3, which wait for the field test of this build.

What changed

Item	Change	Evidence
E7 tests	tests/: test_server.py (289 checks), test_page.mjs (110), check_names.py, run_tests.py. Gate 8 runs them.	Run time 21 s. MOTION_NIGHT_BOOST changed to 14 on purpose: the gate failed with 3 named checks; restored: passed.
E6 old stream endpoint	handle_stream and its route removed, 137 lines.	No caller in the page or the tests. No function became unused through the removal.
E8 manual tiers	handle_focus_set_tier, handle_focus_profiles, their routes, tier_change_kill, manual_override removed, 274 lines.	manual_override was set only by the removed handler and is not saved to disk, so the two conditions that read it reduce to the Adaptive Quality setting alone.
E4 log redaction	One filter on every log handler removes user:password@ and password-style query values. _strip_creds is bounded to the URL's host part.	10 checks: AnyCam lines, %-style arguments, a library logger, the page's log panel, a password that contains '@'.
E1 stage 1	camera_db.py (STREAM_DB, CAMERA_DB; 1,939 lines) and page_script.py (2,782 lines). camera_discovery.py: 18,198 lines in 2.6.8, 13,125 now.	Old and new loaded side by side: both tables equal (76 brands, 40 stream entries), the script equal (124,783 characters), the built page identical (163,326 characters), the same module names. The packaged zip imports from its own folder.

Defect found: probe_http_identity (build plan B19)

The new undefined-name check reads every function's symbol table and reports each global name that no module defines. On its first run it reported four, all one defect.

Fact	Detail
Cause	Commit 2.4.0-rc1.0 added _rtsp_options_fingerprint directly above probe_http_identity and lost the line 'def probe_http_identity(ip, port, timeout)'. The body remains, after a return statement, unreachable. Present in every release since.
Effect 1	run_scan's HTTP identity step raises a name error. A broad except catches it and logs at DEBUG. The step runs only for a device that answers ONVIF discovery and has no card from the port scan. CORRECTED 2026-10-02: this report first said 'on every device'. CrystalHeeler's logs from both systems hold 8 scans and 2,849 DEBUG lines; the step ran 0 times.

Fact	Detail
Effect 2	None. probe_http_for_camera calls the missing function, but its only caller, is_camera_positive, has no caller. CORRECTED 2026-10-02: this report first said step 8 of the camera check fails.
Possible link	Build plan B2, one or two cameras not discovered: only if such a camera answers ONVIF discovery with no open scanned port. Not shown.
This build	Not fixed. Restoring the line changes scan results, and this release is meant to change no behaviour. The four names are listed as known in tests/check_names.py; any new one fails the gate. CrystalHeeler decides (B19).

Tests run

Suite	Result	Scope
Release gate	PASS	Eight gates. Gate 1 compiles each of the 3 files, checks anycam_modules.py against the imports and the Dockerfile (a removed COPY entry failed it, on purpose), 209 top-level definitions, no duplicates. 48 contracts. Gate 8 runs the tests.
Python integration	PASS	289 of 289; 11 new (10 redaction, 1 for the removed endpoints).
JavaScript behaviour	PASS	110 of 110, from the page as built from page_script.py.
Undefined names	NOTED	0 new, 4 known (B19).
Not run	NOTED	No Docker on the build PC: the image was not built, so the new COPY line is checked by the gate's text check and by importing the unpacked zip only. No test covers the scan, the password entry path or most of the snapshot loop, where E6 and E8 removed code.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, Parts 1 to 6, as CLAUDE.md rule 1 requires.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, duplicates, contracts, undefined names	FAIL	probe_http_identity has been truncated since 2.4.0-rc1.0 (B19). Found by this release's new check; five earlier audits passed this row without it.
1.2 Syntax	compile() per file	PASS	Gate 1.
1.4 Async/await	Blocking work on the loop	PASS	Gate 3 clean. The log filter adds two regular expressions per log line, each guarded by a character test.
1.5 Style: line length	Measured added lines	NOTED	New code under 100 characters; moved code unchanged.
1.6 Function design	File size	NOTED	Main file 13,125 lines, down 28%. snap_loop is still about 1,050 lines; stages 2 and 3 are open (E1).
1.7 Mutability	Shared state	NOTED	Unchanged: 66 module-level state objects. Stage 2 needs one shared state module.
1.10 Exceptions	Catches	NOTED	The broad except around the HTTP identity step would hide B19's name error at DEBUG level. The log filter catches only TypeError and ValueError.
1.12 Logging	Credentials	PASS	E4: every handler filtered; tested.
1.16 Refactoring	Dead code	PASS	411 lines of unreachable endpoints and flags removed. One unused function remains from before, handle_storage_page.
1.17 Stdlib first	New tools	PASS	The undefined-name check uses symtable; pyflakes is not installed and was not added.
1.18 Testing	Tests in the repository	PASS	E7. Coverage gaps stated in tests/README.md.
2.x JavaScript	Page script	PASS	Moved, not edited; byte-identical.
5.x Polyglot	Placeholders	PASS	Gate 3 P7 reads the joined files, so it still sees the script's placeholders and build_html's replacements.
6.1 Security	Attack surface, logs	PASS	Three endpoints removed, one of which started a process per request. Credentials removed from logs.

Standard	Check applied	Result	Note
6.2 Dependencies	None added	PASS	Gate 7.
6.3 Version control	Commits and gate	PASS	One commit per item, the gate run before each.

Known issues carried forward

- **B19:** the HTTP identity check does not run. CrystalHeeler decides.
- **E1 stages 2 and 3** are open. Stage 3 needs new tests first.
- The Microseven on the test system A needs a power cycle (B6).
- The Lorex DVR beeping: cause unknown; waiting for the test system B log and the DVR's event entry.
- Skip Non-Reference Frames breaks H.264 cameras (B3).

Acceptance (field test A14)

- The image builds and the add-on starts; the page loads.
- A scan finds the same cameras as 2.6.8.
- Live view, cards, recording and night boost work as in 2.6.8.
- The classic view still steps its quality on its own.
- A log taken after entering a camera password shows no password.

Files changed

File	Change
camera_discovery.py	Tables and script moved out; three endpoints removed; log filter. Version 3.0.0-rc1.0.
camera_db.py, page_script.py, anycam_modules.py	New.
tests/	New: 6 files, 2,612 lines.
verify_release.py	Gate 8; several files; module list checks; 2 new contracts.
package_release.py	New: the packager, in the repository.
Dockerfile	COPY for the two new modules.
docs	Build plan, CLAUDE.md, README; audit tools in docs/audits/tools.
CHANGELOG.md	3.0.0-rc1.0 entry.